Supplementary Information

Evaluation Choices Shape Apparent Quantum Kernel Robustness under Distribution Shift

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Supplementary Results: Scope

This document contains extended results, complete sensitivity tables, and diagnostics supporting the main article. No procedure is defined exclusively here: all data construction, preprocessing, candidate grids, selection rules, inference, external acquisition, geometry calculations, and finite-shot procedures are reported in the main manuscript Methods. Machine-readable values and frozen input hashes are included in the immutable release artifact.

Supplementary Results: Frozen specification curve

The ten rows remain in their prespecified S1–S10 order rather than being ranked by outcome. Every row includes all eight fixed security scenario-groups. Scenario-groups are averaged within EMBER, UNSW-NB15, and ToN-IoT before the three source datasets receive equal weight. The corrected S10 values in Table 1 are identical to the confirmatory endpoint in the main article.

Supplementary Table 1: Frozen specification definitions and source-dataset-equal quantum-minus-classical OOD balanced-accuracy effects. "Oracle" selects and evaluates on the same OOD test labels; "ID-val" uses a disjoint in-distribution validation split. SVC regularization is fixed at $C = 1$ or selected by training-only cross-validation; this axis is not applicable to GPC. S1–S4 use the legacy generation and S5–S10 use v4, so their boundary does not isolate the effect of regularization. Values are descriptive fixed-case summaries, not per-axis causal effects.

ID	Selection	SVC C	Classical reference	Budget	SVC	GPC
S1	OOD oracle	Fixed	Linear/RBF	Native	+0.0313	+0.0340
S2	Legacy ID	Fixed	Linear/RBF	Native	+0.0475	+0.0418
S3	OOD oracle	Fixed	Extended	Native	+0.0058	+0.0145
S4	Legacy ID	Fixed	Extended	Native	+0.0014	+0.0122
S5	OOD oracle	Train CV	Linear/RBF	Native	+0.0034	+0.0125
S6	ID-val	Train CV	Linear/RBF	Native	−0.0041	+0.0059
S7	OOD oracle	Train CV	Extended	Native	−0.0054	+0.0012
S8	ID-val	Train CV	Extended	Native	−0.0038	−0.0029
S9	OOD oracle	Train CV	Extended	Equal	−0.0032	+0.0038
S10	ID-val	Train CV	Extended	Equal	−0.0056	−0.0009

The source-dataset-equal range is 0.0531 for SVC and 0.0447 for GPC. This spread is not an interval: the specifications share data and candidates and deliberately target different estimands.

Supplementary Results: Candidate-budget sampling

Table 2 shows every group–classifier cell under all three prespecified sampling schemes. The primary blocked scheme mirrors the five-dimension block structure of the quantum pool; the other schemes match candidate count but not the same search-space structure. Their close values show that the controlled conclusion is not an artefact of the blocked draw.

Supplementary Table 2: Complete equal-budget sampling sensitivity. Entries are median quantum-minus-classical P1′ OOD differences over 5,000 classical-pool subsamples. “Blocked” is primary; “uniform” and “stratified” are sensitivities. Source-dataset-equal rows average scenario-group medians within the three source datasets. These resampling medians are distinct from the run-level expected-classical endpoint used for the conditional intervals in the main article.

Scenario	Clf.	Blocked	Uniform	Stratified
EMBER $m1$	SVC	−0.0056	−0.0054	−0.0054
	GPC	−0.0004	−0.0009	−0.0013
EMBER $m2$	SVC	−0.0071	−0.0071	−0.0071
	GPC	+0.0007	+0.0008	+0.0005
UNSW-DoS (campaign)	SVC	−0.0036	−0.0034	−0.0036
	GPC	−0.0072	−0.0072	−0.0071
UNSW-DoS (constructed)	SVC	−0.0046	−0.0030	−0.0028
	GPC	−0.0116	−0.0113	−0.0112
UNSW-Recon (campaign)	SVC	+0.0050	+0.0052	+0.0052
	GPC	+0.0046	+0.0047	+0.0049
UNSW-Recon (constructed)	SVC	−0.0090	−0.0089	−0.0089
	GPC	−0.0082	−0.0076	−0.0077
ToN-IoT (campaign)	SVC	−0.0147	−0.0149	−0.0143
	GPC	−0.0095	−0.0094	−0.0096
ToN-IoT (constructed)	SVC	−0.0011	−0.0008	−0.0011
	GPC	+0.0093	+0.0063	+0.0057
Source-dataset-equal	SVC	−0.00576	−0.00555	−0.00548
	GPC	−0.00185	−0.00232	−0.00253

The largest within-cell scheme range is 0.00353 for ToN-IoT constructed under GPC. The next largest is 0.00184 for UNSW-DoS constructed under SVC.

Supplementary Results: Effective-rank matching sensitivities

The predefined rank ratio is $\max(r_Q, r_C)/\min(r_Q, r_C)$. Table 3 reports all calipers for both observational matchers. With replacement, 75.2% of pairs survive the primary 1.25 caliper. One-to-one assignment has a smaller common support and becomes visibly poor without a caliper (95th-percentile ratio 11.45), making the overlap restriction scientifically material.

Supplementary Table 3: Pooled SVC effective-rank matching sensitivity. “Nearest” pairs each quantum candidate to its nearest classical rank with classical reuse; “one-to-one” uses minimum-cost assignment without reuse. Retained percentages use 64,800 possible quantum candidates as denominator. Δ is quantum minus classical OOD balanced accuracy.

Matcher	Caliper	Pairs (%)	Median ratio	95th pct. ratio	Median Δ	Q better
Nearest	1.10	27,928 (43.1)	1.040	1.093	−0.0054	29.4%
	1.25	48,708 (75.2)	1.082	1.226	−0.0056	29.7%
	1.50	60,647 (93.6)	1.112	1.390	−0.0061	28.9%
	2.00	64,408 (99.4)	1.123	1.542	−0.0070	27.8%
	none	64,800 (100)	1.124	1.578	−0.0071	27.6%
One-to-one	1.10	13,205 (20.4)	1.044	1.094	−0.0048	30.2%
	1.25	24,675 (38.1)	1.092	1.232	−0.0053	30.3%
	1.50	35,129 (54.2)	1.151	1.447	−0.0057	30.2%
	2.00	46,898 (72.4)	1.232	1.829	−0.0057	30.8%
	none	64,800 (100)	1.418	11.452	−0.0062	30.9%

At the primary 1.25 caliper, both matchers yield a negative group median in all eight groups.

The result is descriptive of the retained overlap, not a causal family estimate.

Supplementary Results: Geometry associations

Effective-rank association changes sign across regimes, whereas OOD kernel-target alignment is more consistently positive (Table 4). The source-dataset leave-one-out ordering correlations are 0.28 for EMBER, 0.22 for UNSW-NB15, and 0.40 for ToN-IoT when trained on the other two sources.

Supplementary Table 4: Geometry–OOD association and primary nearest-rank pairing for SVC. "Partial" controls kernel family and scale. Matching uses replacement and rank ratio ≤ 1.25; Δ is the median quantum-minus-classical OOD difference. All quantities are observational.

Scenario	ρ rank	Partial	ρ KTA	Matched Δ	Q better
EMBER $m1$	+0.32	+0.36	+0.30	−0.0094	33%
EMBER $m2$	+0.80	+0.82	+0.63	−0.0089	22%
UNSW-DoS (campaign)	+0.44	+0.41	+0.64	−0.0089	27%
UNSW-DoS (constructed)	−0.14	−0.08	+0.39	−0.0057	29%
UNSW-Recon (campaign)	+0.06	+0.06	+0.22	−0.0047	37%
UNSW-Recon (constructed)	−0.15	−0.08	+0.37	−0.0061	30%
ToN-IoT (campaign)	+0.25	+0.27	+0.46	−0.0060	24%
ToN-IoT (constructed)	+0.65	+0.63	+0.59	−0.0013	35%

Supplementary Results: Descriptive oracle comparison

The initial EMBER evaluation selected its representative using OOD test labels and fixed SVC at $C = 1$. Expanding the reference family nearly removes the apparent gain even before target-label-free selection is imposed.

Supplementary Table 5: Descriptive EMBER oracle results under fixed $C = 1$ (18 correlated settings). Mean and median effects are quantum minus classical balanced accuracy. These values are not deployable estimates or independent replicates.

Selection	Classifier	Classical reference	Q wins	Mean Δ	Median Δ
Best-by-OOD	SVC	linear+RBF	17/18	+0.0374	+0.0426
	SVC	extended	15/18	+0.0044	+0.0064
	GP	linear+RBF	15/18	+0.0283	+0.0339
	GP	extended	11/18	+0.0023	+0.0029
Best-by-ID	SVC	linear+RBF	18/18	+0.0775	+0.0794
	SVC	extended	18/18	+0.0149	+0.0152
	GP	linear+RBF	18/18	+0.0649	+0.0664
	GP	extended	18/18	+0.0192	+0.0166

Supplementary Results: Small-cluster interval sensitivity

Table 6 compares the primary conditional cluster-t interval with the prespecified BCa and leave-one-q-split diagnostics. The effective replication unit is five q-split clusters in every row. Differences between interval constructions do not change the qualitative fixed-case message, and an interval containing zero is not interpreted as equivalence.

Supplementary Table 6: Conditional uncertainty sensitivity for all primary equal-budget group–classifier cells. Δ is quantum minus classical OOD balanced accuracy. The cluster-t and 9,999-draw BCa columns are intervals over the same five q-split cluster means ($n = 5$); LOO is the range after leaving out one q-split cluster. All are conditional on the fixed benchmark pools and design, not intervals over a dataset population.

Scenario	Clf.	Δ	95% cluster-t	95% BCa	LOO range
EMBER $m1$	GPC	-0.0009	$[-0.0101, +0.0083]$	$[-0.0062, +0.0054]$	$[-0.0035, +0.0015]$
	SVC	-0.0058	$[-0.0340, +0.0223]$	$[-0.0226, +0.0110]$	$[-0.0120, +0.0014]$
EMBER $m2$	GPC	$+0.0027$	$[-0.0027, +0.0081]$	$[-0.0015, +0.0053]$	$[+0.0017, +0.0043]$
	SVC	-0.0064	$[-0.0137, +0.0009]$	$[-0.0107, -0.0018]$	$[-0.0081, -0.0049]$
UNSW-DoS (campaign)	GPC	-0.0057	$[-0.0120, +0.0005]$	$[-0.0084, +0.0004]$	$[-0.0078, -0.0046]$
	SVC	-0.0033	$[-0.0139, +0.0073]$	$[-0.0101, +0.0029]$	$[-0.0058, -0.0008]$
UNSW-DoS (constructed)	GPC	-0.0108	$[-0.0177, -0.0039]$	$[-0.0141, -0.0054]$	$[-0.0131, -0.0095]$
	SVC	-0.0044	$[-0.0119, +0.0031]$	$[-0.0087, +0.0005]$	$[-0.0067, -0.0026]$
UNSW-Recon (campaign)	GPC	$+0.0052$	$[-0.0019, +0.0123]$	$[-0.0001, +0.0089]$	$[+0.0038, +0.0072]$
	SVC	$+0.0050$	$[-0.0058, +0.0157]$	$[+0.0006, +0.0131]$	$[+0.0011, +0.0065]$
UNSW-Recon (constructed)	GPC	-0.0075	$[-0.0208, +0.0059]$	$[-0.0162, +0.0006]$	$[-0.0106, -0.0040]$
	SVC	-0.0089	$[-0.0146, -0.0031]$	$[-0.0123, -0.0053]$	$[-0.0102, -0.0076]$
ToN-IoT (campaign)	GPC	-0.0090	$[-0.0190, +0.0010]$	$[-0.0166, -0.0034]$	$[-0.0106, -0.0060]$
	SVC	-0.0148	$[-0.0268, -0.0029]$	$[-0.0218, -0.0068]$	$[-0.0180, -0.0129]$
ToN-IoT (constructed)	GPC	$+0.0109$	$[+0.0017, +0.0202]$	$[+0.0060, +0.0178]$	$[+0.0082, +0.0129]$
	SVC	-0.0011	$[-0.0049, +0.0028]$	$[-0.0033, +0.0013]$	$[-0.0019, -0.0003]$

Supplementary Results: External fixed tasks

All 30 prospectively specified task–size–seed units passed the acquisition, split-integrity, nesting, and leakage audits. They contain 37,800 split-level results. Under SVC, size-equal controlled effects for College Scorecard, diabetes readmission, and ACS income are -0.0160, -0.0066, and -0.0131; their protocol contractions are $+0.0393$, $+0.0105$, and $+0.0126$. The task-equal controlled effect is -0.0119 (95% conditional seed-cluster interval $[-0.0205, -0.0033]$, $n = 5$ seeds per task), and contraction is $+0.0208$ ($[+0.0113, +0.0307]$). Under GPC, the task-equal controlled effect is $+0.0077$ ($[-0.0015, +0.0183]$) and contraction is $+0.0324$ ($[+0.0223, +0.0427]$). These fixed-task results corroborate protocol sensitivity, not a universal family ordering.

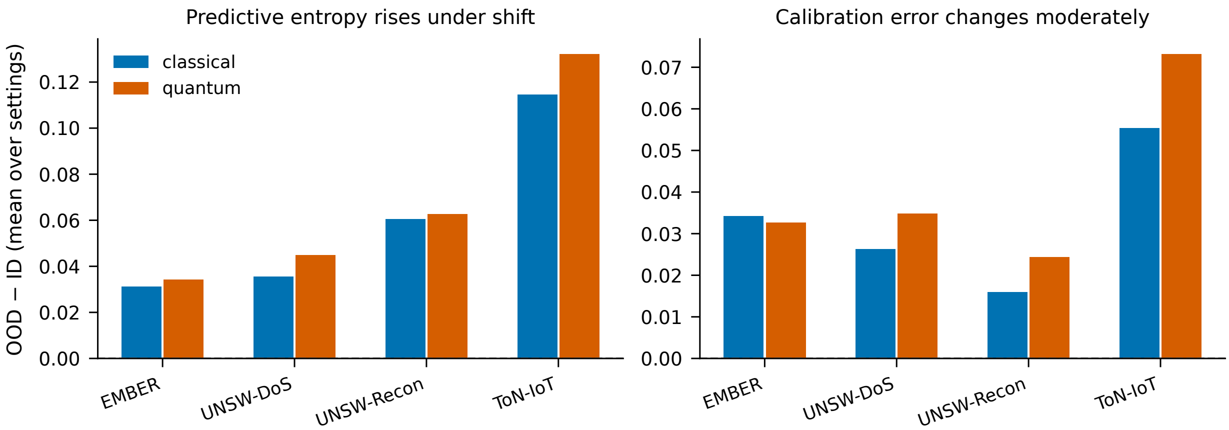
Supplementary Results: Predictive uncertainty

Predictive entropy rises from ID to OOD for both honestly selected families in every scenario-group. Expected calibration error changes moderately and comparably (Figure 1). Because the selected fidelity kernels have no accuracy advantage, these diagnostics show no calibration benefit unavailable to the classical family. Entropy increase alone is not evidence of good calibration.

Supplementary Results: Repeated finite-shot sensitivity

Figure 2 displays all 2,880 finite-shot evaluations: eight fixed exact Gram matrices, four shot counts, 30 measurement replicates, and three PSD-handling conditions. The bands make the conditional Monte Carlo spread visible rather than treating diversity across data configurations as measurement replication. The independent-square heuristic and coherent train-based Nyström extension often narrow or shift the distribution but do not impose a uniform direction. Exact-statevector OOD accuracy is approached unevenly because each sampled training matrix reselects a discrete SVC C.

Across the eight group medians, the perturbed/exact effective-rank ratio declines from 1.93 at 128 shots to 1.13 at 8,192 shots. OOD-KTA under the Nyström extension is estimator-dependent because its OOD square is derived from OOD–train features rather than independently sampled; it is not treated as a mechanism or compared directly with the other KTA estimators. Retained rank, eigenvalue tolerance, blockwise Frobenius changes, negative-spectrum fractions, selected C, and every replicate endpoint are provided in `results/v6/shots_mc/`.



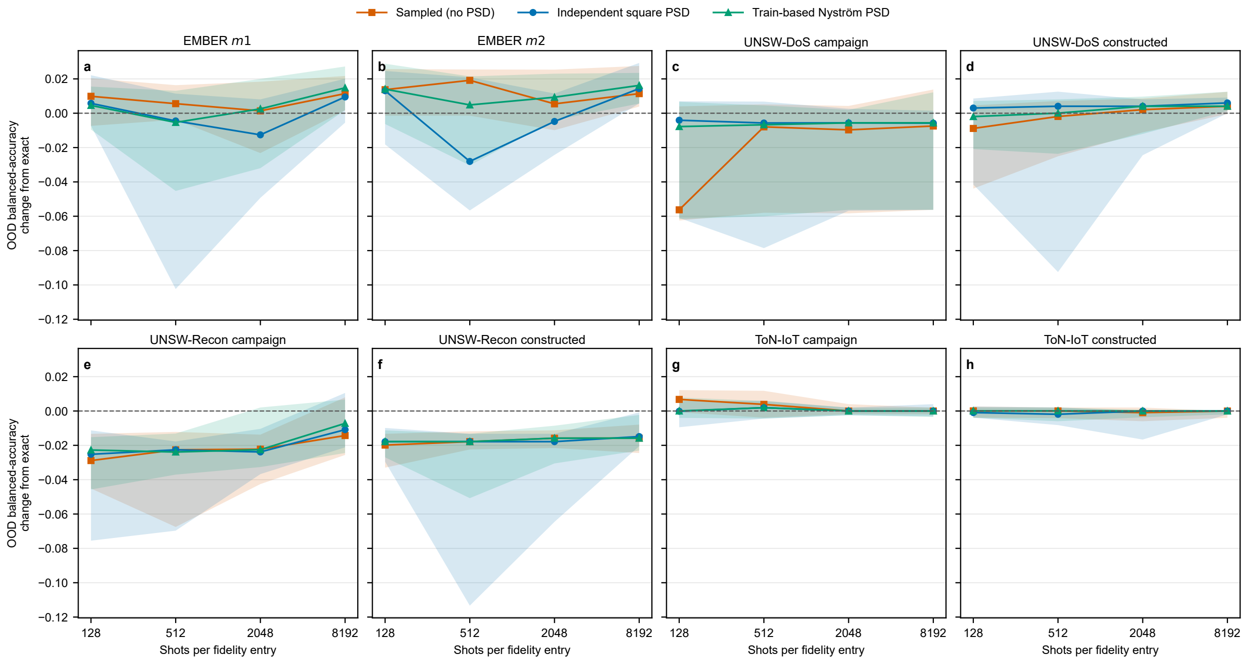
Supplementary Figure 1: Mean ID-to-OOD change in predictive entropy (left) and 15-bin expected calibration error (right) for the P1′-selected configuration of each family. Bars aggregate fixed scenario-groups by source dataset and are descriptive; no population interval is implied.

Supplementary Results: Machine-readable result map

The canonical v0.7.0 release gates are

```
python scripts/analysis/validate_v6_artifacts.py
pytest tests -q
```

The corrected three-source inference, budget schemes, rank sensitivities, and finite-shot outputs are under `results/v6/`; frozen legacy/v4/v5 artifacts remain unchanged. The development repository is `https://github.com/roberto-fernandez-barrios/kernel_shift_framework`, and the concept archive is `https://doi.org/10.5281/zenodo.19147649`. The exact v0.6.0 submission artifact is identified by the reserved immutable version DOI `https://doi.org/10.5281/zenodo.21672470`.



Supplementary Figure 2: Repeated finite-shot OOD change from the exact fixed-configuration endpoint. Panels show one fixed scenario-group Gram matrix. Lines are medians over 30 independently seeded measurement replicates and shaded regions are central 95% Monte Carlo intervals ($n = 30$) conditional on that matrix. Orange squares use the sampled indefinite matrix; blue circles use independent PSD projections of train and OOD squares while leaving rectangular blocks unchanged; green triangles use the coherent train-eigenspace Nyström extension. The model includes fidelity sampling only, not device noise.